I. Exercise 11.9 (Verify the operator norm.)

Firstly, the set $\mathcal{M}(T) = \{M \geq 0 : \forall \mathbf{x} \in \mathbb{F}^n, ||T\mathbf{x}|| \leq M||\mathbf{x}||\}$ clearly has a lower bound 0. Hence by the completeness of $\mathbb{R}$, the infimum indeed exsists.

Now we verify three properties of a norm: positive-definite, positive-homogeneous, and triangle inquality.

If $T = \mathbf{0}$, clearly $||T|| = 0$. Now suppose that $||T|| = 0$. Then $\forall \mathbf{x} \in \mathbb{F}^n$, we have $||T\mathbf{x}|| \leq ||T|| \cdot ||\mathbf{x}|| = 0$. By the positive-definition of vector norm, we have $T\mathbf{x} = 0$. By the arbitariness of $\mathbf{x}$, we have $T = 0$.

The positive-homogeneity follows immediately by $T(\lambda\mathbf{x}) = \lambda T(\mathbf{x})$.

Now for the triangle inequality, We have:

$$||(T+S)\mathbf{x}|| = ||T\mathbf{x} + S\mathbf{x}|| \leq ||T\mathbf{x}|| + ||S\mathbf{x}|| \leq (||T|| + ||S||) \cdot ||\mathbf{x}||, \quad \forall \mathbf{x} \in \mathbb{F}^n.$$

Hence $||T|| + ||S|| \in \mathcal{M}(T+S)$, and hence $||T+S|| = \inf \mathcal{M}(T+S) \leq ||T|| + ||S||$.

II. Exercise 11.3 (Verify d(T,S)=||T-S|| induced a metric space.)

Clearly we have:

- non-negativity: $d(T,S) = ||T-S|| \geq 0$,
- positive-definition: $d(T,S) = 0 \iff T - S = \mathbf{0} \iff T = S$,
- symmetry: $d(T,S) = ||T-S|| = ||S-T|| = d(S,T)$.

Now by the triangle inequality of operator norm, we have:

$$d(T,S) + d(S,P) = ||T-S|| + ||S-P|| \geq ||T-S+S-P|| = ||T-P|| = d(T,P).$$

Hence $d(T,S)$ is indeed a distence, and hence induced a metric space.

III. Exercise 11.16 (Verify the Hilbert-Schmidt norm.)

The positive-definition and the positive-homogeneity are trivial. Now let's verify the triangle inequality.

$$\begin{aligned}
|T+S|^2 &= \sum_{j=1}^{n} ||T\mathbf{e}_j + S\mathbf{e}_j||^2 \\
&= \sum_{j=1}^{n} ||T\mathbf{e}_j||^2 + ||S\mathbf{e}_j||^2 + 2\langle T\mathbf{e}_j, S\mathbf{e}_j\rangle \\
&= |T|^2 + |S|^2 + 2\sum_{j=1}^{n} \langle T\mathbf{e}_j, S\mathbf{e}_j\rangle \\
&\leq |T|^2 + |S|^2 + 2\sqrt{\sum_{j=1}^{n} ||T\mathbf{e}_j||^2 ||S\mathbf{e}_j||^2} \\
&\leq |T|^2 + |S|^2 + 2\sqrt{\sum_{j=1}^{n} ||T\mathbf{e}_j||^2 \sum_{i=1}^{n} ||S\mathbf{e}_i||^2} \\
&= |T|^2 + |S|^2 + 2\sqrt{|T|^2|S|^2} \\
&= |T|^2 + |S|^2 + 2|T| \cdot |S| = (|T| + |S|)^2.
\end{aligned}$$

Hence $|T+S| \leq |T| + |S|$, where the forth step below follows from the Cauchy inequality.

IV. Exercise 11.20 (Prove $|TS| \leq |T||S|$)

By corollary 11.18 we know $||T\mathbf{x}|| \leq |T| \cdot ||\mathbf{x}||$, hence

$$|TS|^2 = \sum_{j=1}^{n} ||TS\mathbf{e}_j||^2 \leq \sum_{j=1}^{n} |T|^2 \cdot ||S\mathbf{e}_j||^2 = |T|^2 \sum_{j=1}^{n} ||S\mathbf{e}_j||^2 = |T|^2|S|^2.$$

V. Exercise 11.36 (Prove $\det(e^X) = e^{\mathbf{tr}X}$)

By Jordan decomposition theorem, $\forall X \in \mathbb{R}^{n\times n}$ or $\mathbb{C}^{n\times n}$, we have

$$X = P^{-1}JP,$$

where P is orthogonal and $J = \mathrm{diag}(J_1, ..., J_k)$ is the Jordan canonical form of X. J_i $(i = 1, ..., k)$ is Jordan block of size m_i, where

$$J_i = \begin{bmatrix} \lambda_i & 1 & & & \\ & \lambda_i & 1 & & \\ & & \ddots & \ddots & \\ & & & \lambda_i & 1 \\ & & & & \lambda_i \end{bmatrix}.$$

Clearly we have:

$$J_i^n = \begin{bmatrix} \lambda_i^n & * & * & \cdots & * \\ & \lambda_i^n & * & \cdots & * \\ & & \ddots & \ddots & \vdots \\ & & & \lambda_i^n & * \\ & & & & \lambda_i^n \end{bmatrix}, \qquad \forall n \in \mathbb{Z}_+,$$

where '*' means an element that we don't care about. And hence

$$e^{J_i} = \begin{bmatrix} \sum_{n=0}^{\infty} \frac{\lambda_i^n}{n!} & * & * & \cdots & * \\ & \sum_{n=0}^{\infty} \frac{\lambda_i^n}{n!} & * & \cdots & * \\ & & \ddots & \ddots & \vdots \\ & & & \sum_{n=0}^{\infty} \frac{\lambda_i^n}{n!} & * \\ & & & & \sum_{n=0}^{\infty} \frac{\lambda_i^n}{n!} \end{bmatrix} = \begin{bmatrix} e^{\lambda_i} & * & * & \cdots & * \\ & e^{\lambda_i} & * & \cdots & * \\ & & \ddots & \ddots & \vdots \\ & & & e^{\lambda_i} & * \\ & & & & e^{\lambda_i} \end{bmatrix}.$$

Note that e^{J_i} is an upper triangle matrix, we can immediately get the determinant:

$$\det(e^{J_i}) = \left(e^{\lambda_i}\right)^{n_i} = e^{n_i\lambda_i}.$$

By lemma 11.32 we have $e^{P_{-1}JP} = P^{-1}e^JP$, hence

$$\det(e^X) = \det(P^{-1}e^JP) = \det(e^J).$$

We could easily verify that for any block diagonal matrix $A = \mathrm{diag}(A_1, ..., A_k)$, we have:

$$e^A = \mathrm{diag}(e^{A_1}, ..., e^{A_k}).$$

And hence

$$\det(e^J) = \det(\mathrm{diag}(e^{J_1}, ..., e^{J_k})) = \prod_{i=1}^{k} \det(e^{J_i}) = \prod_{i=1}^{k} e^{n_i\lambda_i} = e^{\sum_{i=1}^{k} n_i\lambda_i}.$$

Use a well-know conclution in lineaer algebra: tr A is equal to the sum of all eigenvalues of A (containing multiple eigenvalues). We proved that

$$\det(e^X) = e^{\sum_{i=1}^{k} n_i\lambda_i} = e^{\mathrm{tr}\ X}.$$

VI. Exercise 11.50 (Prove the solution of IVP is not overly seneitive to the initial condition)

If $\mathbf{v_0} = \mathbf{w_0}$, then the conclution is trivial. Now we suppose $\mathbf{v_0} > \mathbf{w_0}$. Let $G(t) = \mathbf{v}(t-a) - \mathbf{w}(t-a)$, we assert that $G(t) \geq 0$. Otherwise let

$$t^* = \sup\{t > 0 : \forall 0 \leq s \leq t, G(s) \geq 0\}.$$

By the continuity of G and $G(0) > 0$, also by the assumption that $\exists x$ s.t. $G(x) < 0$, we konw t^* indeed exsists. And clearly $G(t^*) = 0$, hence $\mathbf{v}(t^*) = \mathbf{w}(t^*)$. Then let $\hat{v}$ be the solution of

$$\begin{cases} \frac{\mathrm{d}\hat{v}}{\mathrm{d}t} = f(\hat{v}(t), t + t^*), \\ \hat{v}(0) = \mathbf{v}(t^*). \end{cases}$$

Clearly $\mathbf{v}(t) = \hat{v}(t - t^*)$ and $\mathbf{w}(t) = \hat{v}(t - t^*)$ for any $t > t^*$. By the uniqueness of $\hat{v}$, we have $\mathbf{v}(t) = \mathbf{w}(t)$ for any $v \geq t^*$. That's a contradiction.

Now we proved $G(t)$ $(t \geq 0)$ is a non-negative function which satisfies

$$G'(t) = f(\mathbf{v}(t), t) - f(\mathbf{w}(t), t) \leq L(\mathbf{v}(t) - \mathbf{w}(t)) = LG(t), \quad \forall t \geq 0.$$

Hence by Gronwall's inequality, we have

$$G(t) \leq e^{Lt} G(0).$$

And hence

$$|\mathbf{v}(t) - \mathbf{w}(t)| = \mathbf{v}(t) - \mathbf{w}(t) = G(t - a) \leq e^{L(t-a)} |\mathbf{v}_0 - \mathbf{w}_0|.$$

Similar for $\mathbf{v_0} < \mathbf{w_0}$.

VII. Exercise 11.100 (Compute the first five coefficients C_j)

For the trapezoidal method, we have

$$s = 1,\ \alpha_1 = 1,\ \alpha_0 = -1,\ \beta_1 = \beta_0 = \frac{1}{2}.$$

Hence

$$\begin{aligned}
C_0 &= \alpha_0 + \alpha_1 = 0 \\
C_1 &= -\beta_0 + \alpha_1 - \beta_1 = 0 \\
C_2 &= \frac{1}{2}\alpha_1 - \beta_1 = 0 \\
C_3 &= \frac{1}{6}\alpha_1 - \frac{1}{2}\beta_1 = -\frac{1}{12} \\
C_4 &= \frac{1}{24}\alpha_1 - \frac{1}{6}\beta_1 = -\frac{1}{24}.
\end{aligned}$$

For the midpoint rule, we have

$$s = 2,\ \alpha_2 = 1,\ \alpha_1 = 0,\ \alpha_0 = -1,\ \beta_1 = 2,\ \beta_0 = 0.$$

Hence

$$\begin{aligned}
C_0 &= \alpha_0 + \alpha_1 + \alpha_2 = 0 \\
C_1 &= -\beta_0 + \alpha_1 - \beta_1 + 2\alpha_2 = 0 \\
C_2 &= \frac{1}{2}\alpha_1 - \beta_1 + 2\alpha_2 = 0 \\
C_3 &= \frac{1}{6}\alpha_1 - \frac{1}{2}\beta_1 + \frac{4}{3}\alpha_2 = \frac{1}{3} \\
C_4 &= \frac{1}{24}\alpha_1 - \frac{1}{6}\beta_1 + \frac{2}{3}\alpha_2 = \frac{1}{3}.
\end{aligned}$$

VIII. Exercise 11.102 (Express conditions of $||\mathcal{L}\mathbf{u}(t_n)|| = O(k^3)$)

By lemma 11.95, to achieve $O(k^3)$ accuracy, we must have $C_0 = C_1 = C_2 = 0$, i.e.

$$\sum_{j=0}^{s} \alpha_j = 0, \qquad \sum_{j=0}^{s} j\alpha_j = \sum_{j=0}^{s} \beta_j, \qquad \sum_{j=0}^{s} j^2\alpha_j = 2\sum_{j=0}^{s} j\beta_j,$$

which is equivalanet to

$$\rho(1) = 0, \qquad \rho'(1) = \sigma(1), \qquad \rho'(1) + \rho''(1) = \frac{1}{2}\sigma'(1).$$

IX. Exercise 11.103 (Derive coefficients of LMMs)

Use c++, do symbolic computation with my `fracMatrix`, `fraction` and `bigint` library. The code is here.

```
#include "matrix_fraction.h"
#include <bits/stdc++.h>
using namespace std;

void Adams_Bashforth(){
    cout << "-----------------------Adams Bashforth-----------------------" << endl;
    for(int s = 1; s <= 5; s++){
        cout << "s=" << s << "  p=" << s << "  ";
        fraction a[s+1];
        a[s] = 1;
        a[s-1] = -1;
        fracMatrix coef(s,s);
        fracMatrix rhs(s,1);
        for(int q = 1; q <= s; q++){
            for(int j = 0; j < s; j++){
                coef.element(q-1,j) = fraction(bigPow(j,q-1), bigFact(q-1));
                rhs.element(q-1,0) += fraction(bigPow(j,q), bigFact(q)) * a[j];
            }
            rhs.element(q-1,0) += fraction(bigPow(s,q), bigFact(q)) * a[s];
        }
        fracMatrix b = solve(coef, rhs);
        cout << b.reverse().T();
    }
}

void Adams_Moulton(){
    cout << "------------------------Adams Moulton------------------------" << endl;
    for(int s = 1; s <= 5; s++){
        cout << "s=" << s << "  p=" << s+1 << "  ";
        fraction a[s+1];
        a[s] = 1;
        a[s-1] = -1;
        fracMatrix coef(s+1,s+1);
        fracMatrix rhs(s+1,1);
        for(int q = 1; q <= s+1; q++){
            for(int j = 0; j <= s; j++){
                coef.element(q-1,j) = fraction(bigPow(j,q-1), bigFact(q-1));
                rhs.element(q-1,0) += fraction(bigPow(j,q), bigFact(q)) * a[j];
            }
        }
        fracMatrix b = solve(coef, rhs);
        cout << b.reverse().T();
    }
}

void BDF(){
    cout << "--------------Backward Differentiation Formula---------------" << endl;
    for(int s = 1; s <= 5; s++){
        cout << "s=" << s << "  p=" << s << "  ";
        fracMatrix coef(s+2,s+2);
        fracMatrix rhs = zeros(s+2,1);
        for(int q = 0; q <= s; q++){
            for(int j = 0; j <= s; j++){
                coef.element(q,j+1) = fraction(bigPow(j,q), bigFact(q));
            }
            if(q >= 1) coef.element(q,0) = -fraction(bigPow(s,q-1), bigFact(q-1));
```

```
        }
        coef.element(s+1,s+1) = 1;
        rhs.element(s+1,0) = 1;
        fracMatrix ab = solve(coef, rhs);
        cout << ab.reverse().T();
    }
}

int main(){
    Adams_Bashforth();
    Adams_Moulton();
    BDF();
    return 0;
}
```

You can goto the folder `src` and compile the code by:

```
g++ ex11-103.cpp -o ex -O2
```

And run it by:

```
./ex
```

The running result is here. We computed one more formula than the textbook.

```
-----------------------Adams Bashforth-----------------------
s=1  p=1  [ 1 ]
s=2  p=2  [ 3/2, -1/2 ]
s=3  p=3  [ 23/12, -4/3, 5/12 ]
s=4  p=4  [ 55/24, -59/24, 37/24, -3/8 ]
s=5  p=5  [ 1901/720, -1387/360, 109/30, -637/360, 251/720 ]
-----------------------Adams Moulton-----------------------
s=1  p=2  [ 1/2, 1/2 ]
s=2  p=3  [ 5/12, 2/3, -1/12 ]
s=3  p=4  [ 3/8, 19/24, -5/24, 1/24 ]
s=4  p=5  [ 251/720, 323/360, -11/30, 53/360, -19/720 ]
s=5  p=6  [ 95/288, 1427/1440, -133/240, 241/720, -173/1440, 3/160 ]
--------------Backward Differentiation Formula---------------
s=1  p=1  [ 1, -1, 1 ]
s=2  p=2  [ 1, -4/3, 1/3, 2/3 ]
s=3  p=3  [ 1, -18/11, 9/11, -2/11, 6/11 ]
s=4  p=4  [ 1, -48/25, 36/25, -16/25, 3/25, 12/25 ]
s=5  p=5  [ 1, -300/137, 300/137, -200/137, 75/137, -12/137, 60/137 ]
```

X. Exercise 11.108 (Verify the accuracy of 3-order BDF)

We need to show that

$$\lim_{z\to 1}\frac{\frac{\rho(z)}{\sigma(z)}-\log z}{(z-1)^4}=C\neq 0.$$

By exercise 11.103, we have known that

$$\rho(z)=-\frac{2}{11}+\frac{9}{11}z-\frac{18}{11}z^2+z^3,\quad \sigma(z)=\frac{6}{11}z^3.$$

Use théorem de L'hôpital, we got

$$\begin{aligned}\lim_{z\to 1}\frac{\frac{\rho(z)}{\sigma(z)}-\log z}{(z-1)^4}&=\lim_{z\to 1}\frac{-\frac{1}{3}z^{-3}+\frac{3}{2}z^{-2}-3z^{-1}+\frac{11}{6}-\log z}{(z-1)^4}=\lim_{z\to 1}\frac{z^{-4}-3z^{-3}+3z^{-2}-z^{-1}}{4(z-1)^3}\\&=\lim_{z\to 1}\frac{-4z^{-5}+9z^{-4}-6z^{-3}+z^{-2}}{12(z-1)^2}=\lim_{z\to 1}\frac{20z^{-6}-36z^{-5}+18z^{-4}-2z^{-3}}{24(z-1)}\\&=\lim_{z\to 1}\frac{-120z^{-7}+180z^{-6}-72z^{-5}+6z^{-4}}{24}=-\frac{1}{4}\neq 0.\end{aligned}$$

XI. Exercise 11.109

Prove the necessity first. Suppose that an s-step LMM has order of accuracy p. By lemma 11.95 we have:

$$\mathcal{L}\mathbf{u}(t_n) = \sum_{n=0}^{\infty} C_n k^n \frac{\mathrm{d}^n \mathbf{u}}{\mathrm{d}t^n}(t_n)$$

Since it has order of accuracy p, by definition 11.96 we have $C_0 = C_1 = \cdots = C_p = 0$. If $\mathbf{u}_t$ is a polynomial of degree $< p$, then $\frac{\mathrm{d}^n \mathbf{u}}{\mathrm{d}t^n} \equiv 0$ for any $n \geq q+1$. Hence $\mathcal{L}\mathbf{u} \equiv 0$. However for $\mathbf{u}_t$ of degree p, $\mathcal{L}\mathbf{u} = C_{q+1} k^{q+1} \frac{\mathrm{d}^{q+1}\mathbf{u}}{\mathrm{d}t^{q+1}} \not\equiv 0$.

Now prove the sufficiency. Apply induction to prove $C_0 = \cdots = C_p = 0$, and finally prove $C_{p+1} \neq 0$. Take $\mathbf{u}(t) = c \in \mathbb{C}$, then $\mathbf{u}_t$ is polynomial of degree $< p$, hence

$$\mathcal{L}\mathbf{u} \equiv 0 \implies C_0 c = 0 \ (\forall c \in \mathbb{C}) \implies C_0 = 0.$$

Now suppose we have proved $C_0 = \cdots = C_{m-1} = 0$ for some $m \leq p$. Take $\mathbf{u}(t) = \frac{ct^m}{m!}$, $c \in \mathbb{C}$, then $\mathbf{u}_t$ is polynomial of degree $< p$, hence

$$\mathcal{L}\mathbf{u} = C_m k^m \frac{\mathrm{d}^m \mathbf{u}}{\mathrm{d}t^m} \equiv 0 \implies C_m k^m c = 0 \ (\forall c \in \mathbb{C}) \implies C_m = 0.$$

Hence we proved $C_0 = C_1 = \cdots = C_p = 0$ by induction. Now take $\mathbf{u}(t) = \frac{t^{p+1}}{(p+1)!}$, then $\mathbf{u}_t$ is polynomial of degree p, hence

$$\mathcal{L}\mathbf{u} = C_{p+1} k^{p+1} \frac{\mathrm{d}^{p+1}\mathbf{u}}{\mathrm{d}t^{p+1}} \not\equiv 0 \implies C_{p+1} k^{p+1} \neq 0 \implies C_{p+1} \neq 0.$$

Hence by the definition, the LMM has order of accuracy p.

XII. Exercise 11.113 (Compute the characteristic polynomial of M)

Compute the determinant by the following steps:

$$\begin{aligned}
p_M(z) = \det(zI - M) &= \det \begin{bmatrix} z & -1 & & & \\ & z & -1 & & \\ & & \ddots & \ddots & \\ & & & z & -1 \\ a_0 & a_1 & \cdots & a_{s-2} & z + a_{s-1} \end{bmatrix} \\
&= \det \begin{bmatrix} z & 0 & & & \\ & z & -1 & & \\ & & \ddots & \ddots & \\ & & & z & -1 \\ a_0 & a_1 + a_0 z^{-1} & \cdots & a_{s-2} & z + a_{s-1} \end{bmatrix} \\
&= \cdots \\
&= \det \begin{bmatrix} z & 0 & & & \\ & z & 0 & & \\ & & \ddots & \ddots & \\ & & & z & -1 \\ a_0 & a_1 + a_0 z^{-1} & \cdots & \sum_{j=0}^{s-2} a_j z^{-(s-2-j)} & z + a_{s-1} \end{bmatrix} \\
&= \det \begin{bmatrix} z & 0 & & & \\ & z & 0 & & \\ & & \ddots & \ddots & \\ & & & z & 0 \\ a_0 & a_1 + a_0 z^{-1} & \cdots & \sum_{j=0}^{s-2} a_j z^{-(s-2-j)} & z + \sum_{j=0}^{s-1} a_j z^{-(s-1-j)} \end{bmatrix} \\
&= z^s + z^{s-1} \sum_{j=0}^{s-1} a_j z^{-(s-1-j)} = z^s + \sum_{j=0}^{s-1} a_j z^j
\end{aligned}$$

XIII. Exercise 11.119 (Prove Theorem 11.118 by induction)

We need to prove

$$y_n = \sum_{i=1}^{s-1} \theta_{n-i}\tilde{y}_i + \sum_{i=s}^{n} \theta_{n-i}\psi_i. \tag{1}$$

For $n = 0, ..., s-1$, the equation above becomes

$$y_n = \sum_{i=1}^{s-1} \theta_{n-i}\tilde{y}_i. \tag{2}$$

As described in theorem 11.118, we have

$$\begin{bmatrix} 1 & \theta_1 & \theta_2 & \cdots & \theta_{s-2} & \theta_{s-1} \\ 0 & 1 & \theta_1 & \cdots & \theta_{s-3} & \theta_{s-2} \\ 0 & 0 & 1 & \cdots & \theta_{s-4} & \theta_{s-3} \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & \theta_1 \\ 0 & 0 & 0 & \cdots & 0 & 1 \end{bmatrix} \begin{bmatrix} \tilde{y}_{s-1} \\ \tilde{y}_{s-2} \\ \tilde{y}_{s-3} \\ \vdots \\ \tilde{y}_1 \\ \tilde{y}_0 \end{bmatrix} = \begin{bmatrix} y_{s-1} \\ y_{s-2} \\ y_{s-3} \\ \vdots \\ y_1 \\ y_0 \end{bmatrix}.$$

which implies (2) for $n = 0, 1, ..., s-1$.

Now $\forall m \geq 0$, suppose that (1) is true for $n = 0, ..., m+s-1$, we prove it is also true for $n = m+s$. By the inhomogeneous equation, we have

$$y_{m+s} + \sum_{i=0}^{s-1} \alpha_i y_{m+i} = \psi_{m+s}.$$

And by the homogeneous equation, we have

$$\theta_{m+s} + \sum_{i=0}^{s-1} \alpha_i \theta_{m+i} = 0.$$

Now by the induction assumption, and note that $\theta_{-1} = \cdots \theta_{-s+1} = 0,\ \theta_0 = 1$, we have

$$\begin{aligned} y_{m+s} &= \psi_{m+s} - \sum_{i=0}^{s-1} \alpha_i y_{m+i} \\ &= \psi_{m+s} - \sum_{i=0}^{s-1} \alpha_i \left(\sum_{j=1}^{s-1} \theta_{m+i-j}\tilde{y}_j + \sum_{j=s}^{m+i} \theta_{m+i-j}\psi_j. \right) \\ &= \psi_{m+s} - \sum_{j=1}^{s-1} \tilde{y}_j \sum_{i=0}^{s-1} \alpha_i \theta_{m+i-j} - \sum_{j=s}^{m+s-1} \psi_j \sum_{i=0}^{s-1} \alpha_i \theta_{m+i-j} \\ &= \psi_{m+s} + \sum_{j=1}^{s-1} \tilde{y}_j \theta_{m+s-j} + \sum_{j=s}^{m+s-1} \psi_j \theta_{m+s-j} \\ &= \sum_{j=1}^{s-1} \tilde{y}_j \theta_{m+s-j} + \sum_{j=s}^{m+s} \psi_j \theta_{m+s-j}, \end{aligned}$$

which is (1) for $n = m+s$. Hence we finished the proof by induction.

XIV. Exercise 11.124 (Prove a convergent LMM is consistent)

By lemma 11.122, the convergent LMM is preconsistent, hence

$$C_0 = \sum_{i=0}^{s} \alpha_i = 0.$$

Now we only to prove $C_1 = 0$. Consider the IVP

$$u'(t) = f(t) = 1, \quad u(0) = 0,$$

which has exact solution $u(t) = t$. Now the difference equation gives

$$\sum_{j=0}^{s} \alpha_j U^{n+j} = k \sum_{j=0}^{s} \beta_j. \tag{3}$$

For a convergent method, the initial values satisfie

$$\lim_{k \to 0} U^j = 0, \quad j = 0, ..., s-1. \tag{4}$$

And also

$$\lim_{k \to 0,\ Nk=t} U^N = t \tag{5}$$

Since the convergent LMM must be zero-stable, then $\rho(\zeta)$ does not have a multiple root on the unit circle, hence

$$\rho'(1) = \sum_{j=0}^{s} j\alpha_j \neq 0$$

Let the sequence $\{U^n\}_{n \geq 0}$ be defined by $U^n = Bnk$, where

$$B = \frac{\sum_{j=0}^{s} \beta_j}{\sum_{j=0}^{s} j\alpha_j}. \tag{6}$$

This sequence clearly satisfies (3) as $n = 0$, and satisfies (4). Furthermore, (5) implies that

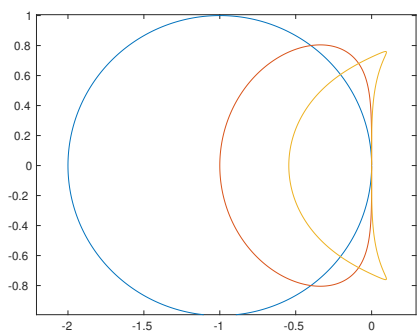
$$t = u(t) = \lim_{k \to 0,\ Nk=t} U^N = \lim_{k \to 0,\ Nk=t} Bnk = Bt,$$

hence we know $B = 1$. Then by (6) we have

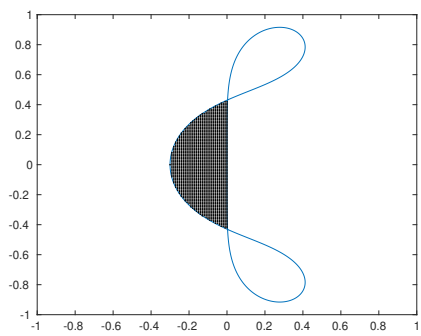
$$C_1 = \sum_{j=0}^{s} j\alpha_j - \sum_{j=0}^{s} \beta_j = 0.$$

XV. Exercise 11.141

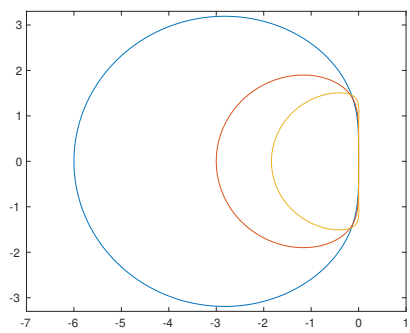
Here are the reproduced figures in Example 11.138, Example 11.139 and Example 11.140.



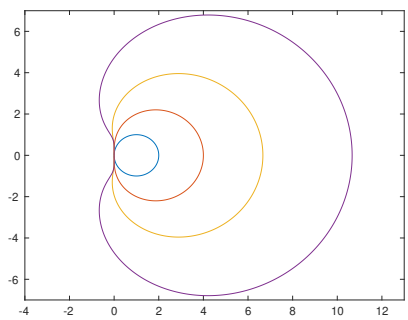
Adams-Bashforth for $p = 1, 2, 3$



Adams-Bashforth for $p = 4$



Adams-Moulton for $p = 3, 4, 5$



BDF for $p = 1, 2, 3, 4$

The figures at Example 11.138 were reproduced by the following matlab code.

```
B = [ 1, 0, 0, 0; -1/2, 3/2, 0, 0; 5/12, -4/3, 23/12, 0; -3/8, 37/24, -59/24, 55/24 ];
% When producing the first figure, replace the 4:4 to 1:3.
for s = 4:4
    b = B(s,:);
    rho = @(x)(x.^s-x.^(s-1));
    sigma = @(x)(b(1)+b(2)*x+b(3)*(x.^2)+b(4)*(x.^3));
    X = 0:0.01:2*pi;
    Y = rho(exp(1i*X))./sigma(exp(1i*X));
    plot(real(Y),imag(Y));
    xlim([-1,1]); ylim([-1,1]);
    hold on
end
% When producing the first figure, the codes below is not necessary.
b = B(4,:);
HX = zeros(201*201, 2); cnt = 0;
for X = -1:0.01:1
    for Y = -1:0.01:1
        k = X + Y*1i;
        rho = [1, -1, 0, 0, 0];
        sigma = [0, b(4), b(3), b(2), b(1)];
        p = rho - k * sigma;
        if max(abs(roots(p)))<=1
            cnt = cnt + 1;
            HX(cnt,:) = [X, Y];
        end
    end
end
scatter(HX(:,1), HX(:,2), 2, "filled", 'k');
```

The figure at Example 11.139 was reproduced by the following matlab code.

```
B = [ -1/12, 2/3, 5/12, 0, 0; 1/24, -5/24, 19/24, 3/8, 0; -19/720, 53/360, -11/30,
323/360, 251/720 ];
for s = 1:3
    b = B(s,:);
    rho = @(x)(x.^(s+1)-x.^(s));
    sigma = @(x)(b(1)+b(2)*x+b(3)*(x.^2)+b(4)*(x.^3)+b(5)*(x.^4));
    X = 0:0.01:2*pi;
    Y = rho(exp(1i*X))./sigma(exp(1i*X));
    plot(real(Y),imag(Y));
    xlim([-7,1]); ylim([-3.3,3.3]);
    hold on
end
```

The figure at Example 11.140 was reproduced by the following matlab code.

```
A = [ 1, -1, 1, 0, 0, 0; 1, -4/3, 1/3, 2/3, 0, 0; 1, -18/11, 9/11, -2/11, 6/11, 0;
1, -48/25, 36/25, -16/25, 3/25, 12/25 ];
for s = 1:4
    b = A(s,:);
    sigma = @(x)(b(2+s)*x.^s);
    X = exp(1i*(0:0.01:2*pi));
    rho = b(1)*(X.^s);
    for i = 2:s+1
        rho = rho + b(i)*(X.^(s+1-i));
    end
    Y = rho./sigma(X);
    plot(real(Y),imag(Y));
    xlim([-4,13]); ylim([-7,7]);
    hold on
end
```